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A TIRE FLANGE AND BEAD SEAT ASSEMBLY FOR A MULTI-PIECE WHEEL RIM

Abstract

The present disclosure provides a tire flange and bead seat assembly for a multi-piece wheel rim. The tire flange and bead seat assembly includes at least one tire flange (101) and at least one bead seat (201). The tire flange (101) includes an end face (103a) and a circumferentially extending tapered radially inward surface (103b). A first tapered surface (203a) includes a taper with angle $\theta_{sub.1}$ configured on a circumferentially extending radially outward surface (205a) of the bead seat and a second tapered surface (203b) includes a taper with angle $\theta_{sub.2}$ configured on a circumferentially extending radially inward surface (205b) of the bead seat (201). The circumferentially extending tapered radially inward surface (103b) includes a taper matching the taper of the first tapered surface (203a) and the circumferentially extending tapered radially inward surface (103b) of the tire flange mounted completely within the first tapered surface (203a) of the bead seat.

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Background/Summary

FIELD

[0001] The embodiments herein generally relate to a multi-piece wheel rim assembly. More particularly, the disclosure relates to a bead seat and a tire flange of a multi piece wheel rim of vehicles used in construction and mining equipment.

BACKGROUND AND PRIOR ART

[0002] Wheels are generally categorized as single piece wheels and multi piece wheels. Multi-piece wheels commonly constitute a wheel rim and a wheel disc, wherein the wheel disc is welded to the wheel rim.

[0003] Generally, in a multi-piece rim, the tire flange of the wheel rim is mounted on the bead seat band. As illustrated in the prior art shown in FIG. 6A and FIG. 6B, a radial clearance is provided between circumferentially extending radially inner surface of the tire flanges (403, 416) and circumferentially extending radially outer surface of the bead seats (412,415) for facilitating mounting of the tire flanges (403, 416). The FIG. 7 clearly illustrates the radial clearance 701 provided between the tire flange and the bead seat. Due to this clearance, a small slippage occurs between the tire flange and the bead seat band during every rotation of the wheel during operation, thereby resulting in fretting related fatigue on both the components. Fretting between the components of the multi-piece wheel rim reduces the life of the components.

[0004] Further, to address the problem of fretting, prior arts disclose a wheel rim wherein the tire flange has increased thickness. Also, the existing tire flange and bead seat assembly of a multi-piece wheel rim includes the radially extending axially outward surface of the tire flange in contact with the radially extending axially inward surface of the bead seat flange upon tire inflation wherein, the tire force acting on the tire flange will induce high stresses at the corners wherein the radially extending axial inward surface of bead seat and circumferentially extending radially outward tapered surface intersect in the respective bead seats. Thereby, the high stress may result in reduction in life of the bead seats.

[0005] Therefore, there is a need for an improved tire flange and bead seat assembly of a multi-piece wheel rim for preventing fretting. Moreover, there is a need for a tire flange and bead seat assembly with reduced stresses and imparting improved life of the multi-piece wheel rim.

OBJECTS

[0006] Some of the objects of the present disclosure are described herein below:

[0007] The main objective of the present disclosure is to provide a tire flange and bead seat assembly of a multi-piece wheel rim.

[0008] Another objective of the present disclosure is to provide a tire flange and bead seat assembly of a multi-piece wheel rim for preventing fretting.

[0009] Still another objective of the present disclosure is to provide a tire flange and bead seat assembly for reducing stress and enhancing rigidity on the bead seat band of a multi piece wheel rim

[0010] Yet another objective of the present disclosure is to provide a tire flange and bead seat assembly for reducing weight of the wheel rim by providing a tire flange with reduced thickness.

[0011] The other objectives and advantages of the present disclosure will be apparent from the following description when read in conjunction with the accompanying drawings, which are incorporated for illustration of preferred embodiments of the present disclosure and are not

intended to limit the scope thereof.

SUMMARY

[0012] In view of the foregoing, an embodiment herein provides a tire flange and bead seat assembly of a multi-piece wheel rim.

[0013] In accordance with an embodiment, the multi-piece wheel rim comprises at least one tire flange and at least one bead seat. The tire flange includes an end face and a circumferentially extending tapered radially inward surface. The bead seat includes a first tapered surface including a taper with angle of inclination $\theta_{\text{sub.1}}$ configured on a circumferentially extending radially outward surface and a second tapered surface including a taper with angle of inclination $\theta_{\text{sub.2}}$ configured on a circumferentially extending radially inward surface. The circumferentially extending tapered radially inward surface of the tire flange is provided with a taper matching the taper of the first tapered surface and the circumferentially extending tapered radially inward surface of the tire flange is mounted completely within the first tapered surface of the bead seat upon tire inflation.

[0014] In accordance with an embodiment, the tire flange includes a radially extending axially outward surface and a radially extending axially inward surface, wherein the radially extending axially outward surface is divided into an upper surface and a lower surface. A thickness T1 is formed between the upper surface and the radially extending axially inward surface and a thickness T2 is formed between the lower surface and the radially extending axially inward surface, wherein the thickness T2 is less than the thickness T1 for providing a clearance between the lower surface and a radially extending axially inward flange surface of a bead seat flange.

[0015] In accordance with an embodiment, an outer fillet is provided at an edge between a radially extending axially outward surface of the tire flange and the circumferentially extending tapered radially inward surface and an inner fillet is provided at the edge between a radially extending axially inward surface of the tire flange and the circumferentially extending tapered radially inward surface for reducing stress concentration on the tire flange.

[0016] In accordance with an embodiment, the angle of inclination $\theta_{\text{sub.1}}$ is greater than the angle of inclination $\theta_{\text{sub.2}}$ for gradually increasing section in critical region of the bead seat under load, thereby enhancing rigidity of the bead seat and restricting the tire flange in not establishing contact with a radially extending axially inward flange surface of the bead seat upon tire inflation.

[0017] In accordance with an embodiment, the thickness T1 and T2 of the tire flange is determined based on load acting on the wheel, thereby reducing weight of the wheel.

[0018] These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following description, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] The detailed description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

[0020] FIG. 1 illustrates a sectional view of a tire flange of the multi-piece wheel rim, according to an embodiment herein;

[0021] FIG. 2 illustrates a sectional view of a bead seat of the multi-piece wheel rim, according to

an embodiment herein;

[0022] FIG. **3** illustrates a tire flange and bead seat assembly of the multi-piece wheel rim, according to an embodiment herein;

[0023] FIG. **4** illustrates a multi piece wheel rim assembly showing rim base, saddle ring, endless ring, bead seat and tire flange, according to an embodiment herein;

[0024] FIG. **5a** illustrates exemplary stresses acting on bead seat of a multi piece wheel rim, according to prior art;

[0025] FIG. **5b** illustrates stresses acting on the bead seat of the multi piece wheel rim, according to an embodiment herein;

[0026] FIG. **6a** illustrates a multi piece wheel rim assembly with saddle lock ring according to prior art;

[0027] FIG. **6b** illustrates a multi piece wheel rim assembly with conventional split lock ring, according to prior art; and

[0028] FIG. **7** illustrates a conventional multi piece wheel rim assembly showing radial clearance between tire flange and bead seat band, according to prior art.

LIST OF NUMERALS

[0029] **101**—Tire flange [0030] **101a**—Radially extending axially inward surface [0031] **101b**—Radially extending axially outward surface [0032] **102**—Transition point [0033] **102a**—Upper surface [0034] **102b**—Lower surface [0035] **103a**—End face [0036] **103b**—Circumferentially extending tapered radially inward surface [0037] **104a**—Outer fillet [0038] **104b**—Inner fillet [0039] **201**—Bead seat band [0040] **203a**—First tapered surface [0041] **203b**—Second tapered surface [0042] **205a**—Circumferentially extending radially outward surface [0043] **205b**—Circumferentially extending radially inward surface [0044] **207**—Bead seat flange [0045] **207a**—Radially extending axially inward flange surface [0046] **401**—Rim [0047] **403**—Gutter region [0048] **405**—Saddle ring [0049] **407**—Endless ring [0050] **70**—Radial clearance

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0051] The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.

[0052] As mentioned above, there is a need for an improved tire flange and bead seat assembly of a multi-piece wheel rim for preventing fretting. In particular, there is a need for a tire flange and bead seat assembly with reduced stresses and imparting improved life of the multi-piece wheel rim. The embodiments herein achieve this by providing “A tire flange and bead seat assembly for a multi-piece wheel rim”. Referring now to the drawings and more particularly to FIG. **1** to FIG. **4** where similar reference characters denote corresponding features consistently throughout the figures, there are shown preferred embodiments.

[0053] FIG. **1** illustrates a sectional view of a tire flange of the multi-piece wheel rim. The tire flange **101** includes an end face **103a** and a circumferentially extending tapered radially inward surface **103b**, a radially extending axially outward surface **101b**, and a radially extending axially inward surface **101a**. In an embodiment, a transition point **102** divides the radially extending axially outward surface **101b** of the tire flange into an upper surface **102a** and a lower surface **102b**. The tire flange includes a first thickness **T1** formed between the radially extending axially inward surface **101a** and the upper surface **102a**, and a second thickness **T2** formed between the radially extending axially inward surface **101a** and the lower surface **102b**. In an embodiment, the first thickness **T1** is greater than the second thickness **T2**.

[0054] In an embodiment, the circumferentially extending tapered radially inward surface **103b** is

tapered for matching a taper of a bead seat. The circumferentially extending tapered radially inward surface **103b** is a circumferentially extending radially inward surface of the tire flange **101** including a single taper.

[0055] In an embodiment, the circumferentially extending tapered radially inward surface **103b** includes an outer fillet **104a**, and an inner fillet **104b**. The outer fillet **104a** is provided at the edge between the radially extending axially outward surface **101b** and the circumferentially extending tapered radially inward surface **103b** and the inner fillet **104b** is provided at the edge between the radially extending axially inward surface **101a** and the circumferentially extending tapered radially inward surface **103b** for preventing stress concentration on the tire flange.

[0056] FIG. 2 illustrates a sectional view of a bead seat of the multi-piece wheel rim. In an embodiment, the bead seat **201** includes a circumferentially extending radially outward surface **205a** and a circumferentially extending radially inward surface **205b**. The circumferentially extending radially outward surface **205a** includes a first tapered surface **203a**. The first tapered surface **203a** is provided for supporting the tire flange of the wheel rim. The circumferentially extending radially inward surface **205b** includes a second tapered surface **203b**.

[0057] The bead seat **201** includes a bead seat flange **207**, wherein the bead seat flange **207** includes a radially extending axially inward flange surface **207a**.

[0058] In an embodiment, the first tapered surface **203a** is inclined at an angle $\theta_{\text{sub.1}}$ relative to the horizontal. In an embodiment, the second tapered surface **203b** is inclined at an angle $\theta_{\text{sub.2}}$ relative to the horizontal. The angle $\theta_{\text{sub.1}}$ of the first tapered surface is greater than the angle of inclination $\theta_{\text{sub.2}}$ of the second tapered surface.

[0059] FIG. 3 illustrates a tire flange and bead seat assembly of the multi-piece wheel rim. In an embodiment, the circumferentially extending tapered radially inward surface **103b** of the tire flange **101** is mounted on the first tapered surface **203a** of the bead seat. In an embodiment, the lower surface **102b** of the tire flange forms a clearance C relative to the radially extending axially inward flange surface **207a** upon tire inflation, thereby preventing contact between the tire flange **101** and the bead seat flange **207**.

[0060] Dimensions of the first tapered surface **203a** and the second tapered surface **203b** of the bead seat **201** are determined based on load acting by tire forces to ensure rigidity of bead seat band wherein engagement of the tire flange **101** is restricted to the first tapered surface **203a** of the bead seat and does not establish contact with the radially extending axially inward flange surface **207a** of the bead seat band.

[0061] In an embodiment, the taper of the circumferentially extending tapered radially inward surface **103b** of the tire flange matches the taper of the first tapered surface **203a** of the bead seat. The circumferentially extending tapered radially inward surface **103b** is completely engaged within the first tapered surface **203a** of the bead seat.

[0062] FIG. 4 illustrates a multi piece wheel rim assembly showing rim base, saddle ring, endless ring, bead seat and tire flange, according to an embodiment. The wheel includes a rim **401**, a gutter region **403**, a saddle ring **405** and an endless ring **407**. The circumferentially extending tapered radially inward surface **103b** of the tire flange **101** is mounted on the bead seat **201** at the first tapered surface **203a**. Upon tire inflation, the circumferentially extending tapered radially inward surface **103b** is completely mounted within the first tapered surface **203a** thereby the tire flange **101** not establishing contact with the radially extending axially inward flange surface **207a** of the bead seat **201** and preventing fretting. In an embodiment, a tire force A from the tire flange acts on the first tapered surface **203a** of the bead seat, wherein the tire force A is supported by reactions B and C acting on the second tapered surface **203b** of the bead seat. As the tire flange **101** and bead seat **201** assembly, restricts engagement of the tire flange **101** within the first tapered surface **203a** of the bead seat, the bead seat acts a simply supported beam, thereby reducing stresses induced in the bead seat and enhancing the life of both the tire flange and bead seat of the multi-piece wheel rim.

[0063] In an embodiment, the angle of inclination $\theta_{sub.2}$ of the second tapered surface **203b** being lesser than the angle of inclination $\theta_{sub.1}$ of the first tapered surface imparts a gradually increasing section in the critical region under the load of the bead seat and enhances rigidity of the bead seat **201**.

[0064] FIG. 5a illustrates exemplary stresses acting on a prior art bead seat of a multi piece wheel rim, designed for use in a 57 inch wheel of a 240T mining dump truck. FIG. 5b illustrates stresses acting on the bead seat of the multi piece wheel rim, according to an embodiment, wherein the multi-piece wheel rim is designed for use in the same 57 inch wheel of 240T mining dump truck. As illustrated, values of stresses of bead seat are lower approximately by 40% as against the stress levels induced in the bead seat of prior art.

[0065] A main advantage of the present disclosure is that the tire flange and bead seat assembly of a multi-piece wheel rim prevents fretting.

[0066] Another advantage of the present disclosure is that the tire flange and bead seat assembly of a multi-piece wheel rim reduces stresses acting on the bead seat.

[0067] Still another advantage of the present disclosure is that the tire flange and bead seat assembly of a multi-piece wheel rim enhances rigidity of the bead seat band.

[0068] Yet another advantage of the present disclosure is that the tire flange and bead seat assembly facilitates utilization of a tire flange with reduced thickness, thereby reducing weight of the wheel rim.

[0069] Still another advantage of the present disclosure is that the tire flange and bead seat assembly increases life of components of the multi-piece wheel rim.

[0070] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the embodiments as described herein.

Claims

1. A tire flange and bead seat assembly of a multi-piece wheel rim, comprising: at least one tire flange (**101**); at least one bead seat (**201**); characterized in that wherein the tire flange (**101**) including an end face (**103a**) and a circumferentially extending tapered radially inward surface (**103b**); a first tapered surface (**203a**) including a taper with angle of inclination $\theta_{sub.1}$ configured on a circumferentially extending radially outward surface (**205a**) of the bead seat (**201**); a second tapered surface (**203b**) including a taper with angle of inclination $\theta_{sub.2}$ configured on a circumferentially extending radially inward surface (**205b**) of the bead seat (**201**); the circumferentially extending tapered radially inward surface (**103b**) of the tire flange provided with a taper matching the taper of the first tapered surface (**203a**); and the circumferentially extending tapered radially inward surface (**103b**) of the tire flange mounted completely within the first tapered surface (**203a**) of the bead seat upon tire inflation, thereby preventing fretting.
2. The tire flange and bead seat assembly as claimed in claim 1, wherein the tire flange (**101**) including a radially extending axially outward surface (**101b**) and a radially extending axially inward surface (**101a**), wherein the radially extending axially outward surface (**101b**) divided into an upper surface (**102a**) and a lower surface (**102b**); wherein a thickness T1 formed between the upper surface (**102a**) and the radially extending axially inward surface (**101a**) and a thickness T2

formed between the lower surface (102b) and the radially extending axially inward surface (101a); and wherein the thickness T2 less than the thickness T1 for providing a clearance between the lower surface (102b) and a radially extending axially inward flange surface (207a) of a bead seat flange (207).

3. The tire flange and bead seat assembly as claimed in claim 1, wherein an outer fillet (104a) provided at an edge between a radially extending axially outward surface (101b) of the tire flange (101) and the circumferentially extending tapered radially inward surface (103b) and an inner fillet (104b) provided at the edge between a radially extending axially inward surface (101a) of the tire flange (101) and the circumferentially extending tapered radially inward surface (103b) for reducing stress concentration on the tire flange (101).

4. The tire flange and bead seat assembly as claimed in claim 1, wherein the angle of inclination $\theta_{sub.1}$ greater than the angle of inclination $\theta_{sub.2}$ for gradually increasing section in critical region of the bead seat (201) under load, thereby enhancing rigidity of the bead seat and restricting the tire flange (101) in not establishing contact with a radially extending axially inward flange surface (207a) of the bead seat (201) upon tire inflation.

5. The tire flange and bead seat assembly as claimed in claim 2, wherein the thickness T1 and T2 of the tire flange (101) determined based on load acting on the wheel, thereby reducing weight of the wheel.
